UDACITY
Part of Accenture

Sleep Health and Lifestyle Analysis

Descriptive Statistics Analysis

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Agenda

Results of sleep health and lifestyle analysis

- Data Description
- Typical Amount of Physical Activity
- Number of Daily Steps
- Distribution of Heart Rates

Data Description

Sleep Duration is an example of a continuous variable in the dataset.

- → Values like 5.9, 6.1, 7.8 hours; can take any decimal value in a range.

- Daily Steps is an example of an integer variable in the dataset.
- → Recorded as whole numbers (e.g., 4200, 10000, 3000 steps).

- Quality of Sleep is an example of an ordinal categorical variable in the dataset.
- → Rating scale 1-10 with natural ordering; higher = better sleep quality.

- Occupation is an example of a nominal categorical variable in the dataset.
- → Categories (Doctor, Engineer, Teacher) with no inherent ordering.

Typical Amount (Minutes) of Physical Activity

Measures of Center for Physical Activity Level (minutes/day):

- Mean: 59.17 minutes/day
- Median: 60.0 minutes/day
- Mode: 60.0 minutes/day

- **Skewness Analysis:**
- The mean (59.17) is slightly less than the median (60.0),
- indicating a slightly left-skewed (negatively skewed) distribution.
- This suggests a small tail of lower physical activity values pulling
- the mean below the median.

Analysis of Daily Steps Taken

Measures of Spread for Daily Steps:

- Standard Deviation: 1,617.92 steps
- Maximum: 10,000 steps
- Minimum: 3,000 steps
- Range: 7,000 steps

- **Additional Measures of Spread:**
- Q1 (25th percentile): 5,600 steps
- Q3 (75th percentile): 8,000 steps
- IQR (Q3 − Q1): 2,400 steps
- Variance: 2,617,651.14

Distribution of Heart Rates

Distribution Shape: Right-skewed (positively skewed)

- Mean (70.17 bpm) > Median (70.0 bpm), confirming right skew.

- **Outliers (IQR method, bounds: 62–78 bpm):**
- Values above 78 bpm: 80, 81, 82, 83, 84, 85, 86 bpm
- These 7 high heart rate values may indicate individuals with
- elevated cardiovascular stress or sleep disorders.

